MICROLOG

Technical datasheet

MICROLOG-01 · MICROLOG-01R · Revision 0014 · November 2026





GUBELLINI s.a.s. di Diego Gubellini & C. – Via Euridia Bergianti 10B, 40059 Medicina (BO) Italy
P.IVA / VAT IT03466001207 · info@gubellinielectronics.com · Tel. +39 347 9200208 · WhatsApp +39 380 1978288
www.gubellinielectronics.com

Contents

1. Overview 2
1.1 Description 2
1.2 Main features 2
2. Versions 3
2.1 Model comparison 3
2.2 Lightweight version and rugged version (R) 3
2.3 Expansion with CAN-BOARD and GUBENAV 3
3. Technical data 5
3.1 Channels and sampling 5
3.2 CAN bus 5
3.3 Memory (microSD) 5
3.4 USB-C 5
3.5 Power supply 6
3.6 AMP Super Seal 1.5 connector 6
3.7 Operating environment 6
3.8 Dimensions and weight 6
4. Typical connections 8
5. Configuration with GUBELLINI DataStudio 9
5.1 The software 9
5.2 First connection 9
5.3 CAN channels 10
5.4 Auto-start conditions 10
5.5 Saving and sending the configuration 10
5.6 Datasets and download from the microSD 10
5.7 AI assistant 11
6. microSD recording time 12
7. Ordering information 13
7.1 Data loggers 13
7.2 CAN modules 13
7.3 Related products 13

1 Overview

1.1 Description

MICROLOG is a professional, compact and easy-to-use CAN data logger, designed for many fields: automotive, motorsport, laboratory analysis and monitoring, machinery monitoring, precision agriculture, industrial monitoring and process analysis. It records up to **16 signals** read from the CAN bus, with the OBD-II, SAE J1939, ISOBUS and Open CAN protocols, on a microSD.

Configuration is done with the free **GUBELLINI DataStudio** software via USB-C; data is downloaded by reading the microSD on the PC and analysed with the free **GUBELLINI DataAnalysis** software (chap. 5). The **CAN-BOARD** boards and the **GUBENAV** module add accelerometers, gyroscopes, GPS position, load cells, optoisolated inputs and other measurements (§ 2.3).

1.2 Main features

- **16 CAN channels** recorded simultaneously (CAN1-CAN16).
- **Sampling rate** programmable per channel from **0.01 Hz to 200 Hz**: suitable both for long-term monitoring and for dynamic analysis.
- High-speed **CAN 2.0B**, 11- or 29-bit IDs: OBD-II, SAE J1939, ISOBUS (ISO 11783), CANopen (EN 50325-4) or custom signals, also from DBC files.
- **Auto-start**: recording from power-on or when a condition on a CAN channel occurs.
- **microSD** from 16 to 128 GB, read directly by the PC.
- **USB-C** as a USB HID device: no driver to install.
- **IPX7**, from **-20 °C to +85 °C**, 4-way **AMP Super Seal 1.5** automotive connector.

2 Versions

2.1 Model comparison

MICROLOG is available in two versions: the lightweight version and the rugged **R** version. The electronics and all technical characteristics are identical: only the housing differs.

	MICROLOG-01	**MICROLOG-01R**
Order code	GE-ML-0001	On request
Housing	Nylon PA12 + 35% GF	Extruded anodised aluminium
Recommended use	Where size and weight matter	Harsh environments, maximum mechanical strength
Channels, CAN, microSD, USB-C	Identical (chap. 3)	

2.2 Lightweight version and rugged version (R)



Figure 1 – MICROLOG-01: PA12 + 35% GF housing.



Figure 2 – MICROLOG-01R: extruded anodised aluminium.

- **Lightweight version**: housing in PA12 nylon filled with 35% glass fibre, a material with excellent mechanical properties, resistant to shocks and vibration.
- **R version**: extruded anodised aluminium housing, for maximum mechanical strength and long-term durability in the harshest environments.

2.3 Expansion with CAN-BOARD and GUBENAV

MICROLOG records everything that travels on the CAN bus: connecting the **CAN-BOARD** boards or the **GUBENAV** module to the bus adds new measurements, recorded as CAN channels.

Module	Measurement transmitted on the CAN
GUBENAV	High-dynamics GPS position and inertial (IMU) data, up to 100 Hz
CAN-BOARD-ACCELEROMETER	3-axis accelerations
CAN-BOARD-GNSS	Latitude, longitude, ground speed, date and time
CAN-BOARD-LOADCELL	Load cell
CAN-BOARD-OPTO	Frequency of 2 optoisolated digital inputs
CAN-BOARD-THERMOK	Type K thermocouple

The boards are configured from DataStudio with the **Sensor configuration** module; their CAN IDs are then entered in the MICROLOG channels.

3 Technical data

3.1 Channels and sampling

	Value
Channels	16 CAN channels (CAN1-CAN16) recorded simultaneously
Sampling rate	From 0.01 Hz to 200 Hz, programmable for each channel
Conversion	Label, unit of measurement, multiplier and offset for each channel, set from DataStudio
Recording start	*Always On* (at power-on) or *Auto-start* (when a CAN channel goes above, below or equals a threshold); the dataset is closed at power-off

3.2 CAN bus

	Value
Transceiver	High-Speed CAN 2.0B
Bus speed	1 Mbps, 500 kbps, 250 kbps
ID format	11-bit (standard) or 29-bit (extended)
Protocols	OBD-II, SAE J1939, ISOBUS (ISO 11783), CANopen (EN 50325-4), CUSTOM signals (also from DBC files)
OBD-II	The logger sends the requests on the bus and records the ECU responses
J1939	The logger listens to the bus and filters the messages by the PGN contained in the 29-bit ID
Termination	120 Ω resistor integrated in the data logger

3.3 Memory (microSD)

	Value
Media	microSD in RAW format (unformatted); data is organised in datasets
Supported capacities	16, 32, 64 and 128 GB (higher capacities work but are not fully used)
Reading	By removing the microSD and reading it on the PC in DataStudio

3.4 USB-C

USB-C connector on the side of the housing, protected by a rubber cap. MICROLOG appears to the PC as a **USB HID** device, with no driver to install. Via USB you send and receive the configuration and read the firmware version and the list of recorded datasets; the data is downloaded from the microSD.

3.5 Power supply

	Value
Supply voltage	From 7 to 24 V DC (connector pin 1)

3.6 AMP Super Seal 1.5 connector

Power and CAN bus pass through a single 4-way **AMP Super Seal 1.5 series** automotive connector (part number 282106-1), watertight, which maintains the logger's degree of protection.

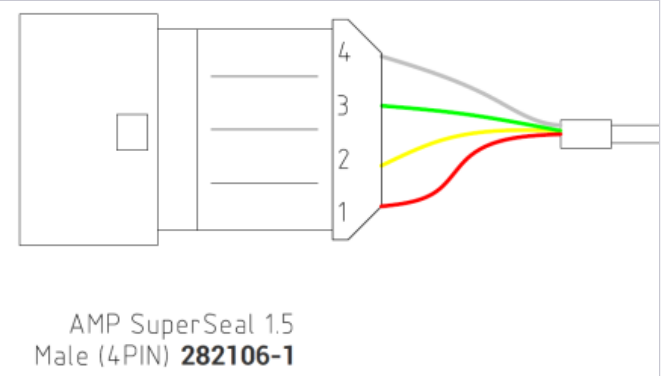


Figure 3 – 4-pin male AMP Super Seal 1.5 connector (282106-1).

Pin	Signal	Type
1	VBatt (7-24 V DC)	Power supply
2	CAN H	CAN bus
3	CAN L	CAN bus
4	GND	Ground

3.7 Operating environment

	Value
Degree of protection	IPX7
Operating temperature	From -20 °C to +85 °C

3.8 Dimensions and weight

	MICROLOG-01	MICROLOG-01R
Housing	PA12 + 35% GF	Extruded anodised aluminium
Length × width × height	66.2 × 38 × 27 mm	69.5 × 35 × 26 mm
Weight	180 g	On request

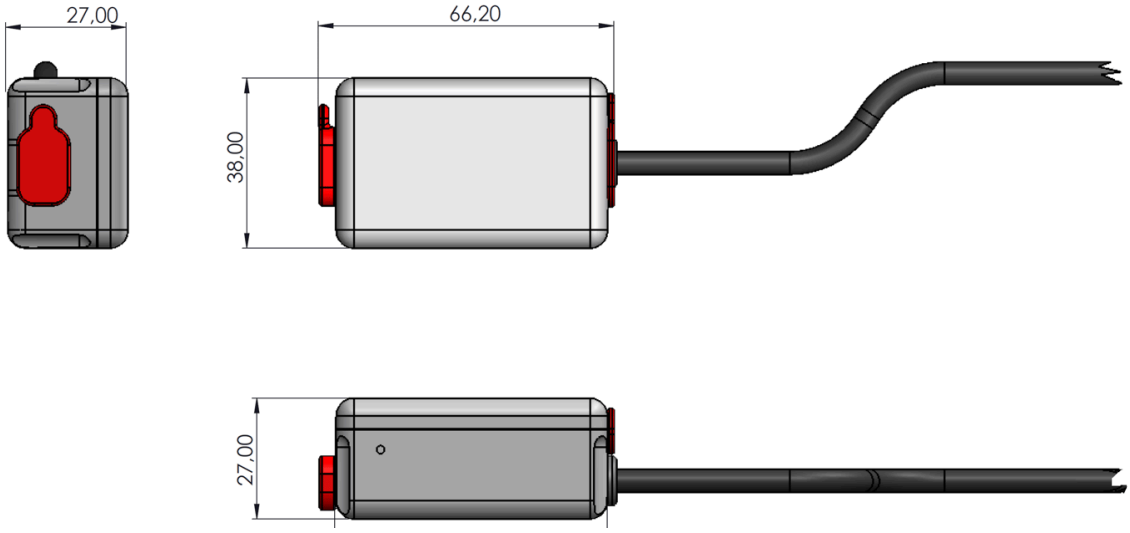


Figure 4 – MICROLOG-01 dimensions (mm).

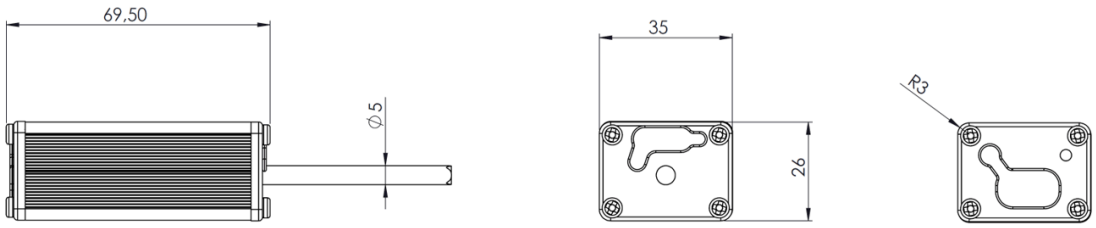


Figure 5 – MICROLOG-01R dimensions (mm).

4 Typical connections

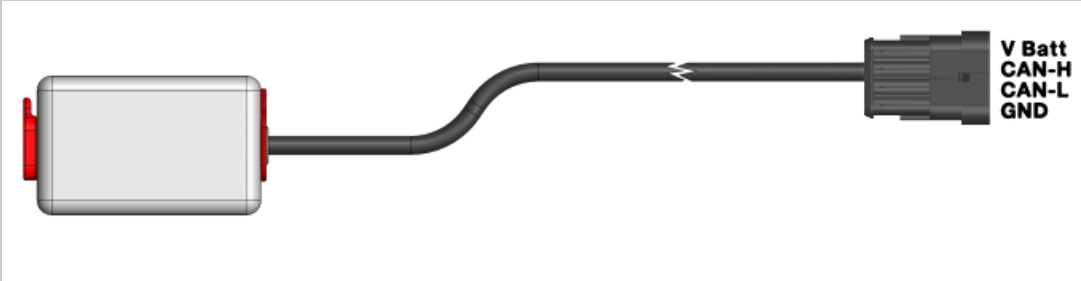


Figure 6 – Wiring diagram: power supply and CAN line on the 4-pin connector.

Connection	MICROLOG pins	Notes
Power supply	1 (+), 4 (GND)	From 7 to 24 V DC, for example from the vehicle battery via the ignition switch.
Vehicle or machine CAN line	2 (CAN H), 3 (CAN L)	120 Ω termination integrated in the logger.
CAN-BOARD and GUBENAV	2 (CAN H), 3 (CAN L), 4 (GND)	Same CAN line: each module transmits with its own IDs, at the chosen bus speed.

OBD-II diagnostic socket (J1962)

OBD-II socket pin	Signal	MICROLOG pin
16	Battery power	1
6	CAN High	2
14	CAN Low	3
4 / 5	Chassis ground / signal ground	4

5 Configuration with GUBELLINI DataStudio

5.1 The software

Software	What it does
GUBELLINI DataStudio (free)	Configures Gubellini Electronics devices (MICROLOG, GUBELOG, GUBENAV, CAN boards); imports signals from DBC files; downloads datasets from the microSD and exports to MoTeC, AEM and CSV; manages datasheets and part counters (hours and km); includes an AI assistant for device connection and configuration.
GUBELLINI DataAnalysis (free)	Analyses the downloaded datasets: plots versus time or distance, layouts with multiple plots, math channels, track map, import and export of on-board video; it includes an **AI assistant** that analyses the data, writes reports and generates the code of math channels.

The DataStudio user manual, included in the program (**Help → User manual (PDF)**), describes every function in detail; the main steps for MICROLOG are summarised here.

5.2 First connection

1. Install DataStudio and connect MICROLOG to the PC with the USB-C cable. No driver is needed.
2. In the main window press **MICROLOG**: the Micrologger module opens. The status bar shows *Connected* or *Disconnected*, checked every second.
3. **Micrologger → Read configuration** loads the logger's configuration; **Micrologger → Firmware version** shows its firmware.

> **One USB HID device at a time**
>
> MICROLOG and the CAN boards appear to the PC in the same way: connect only one HID device at a time.

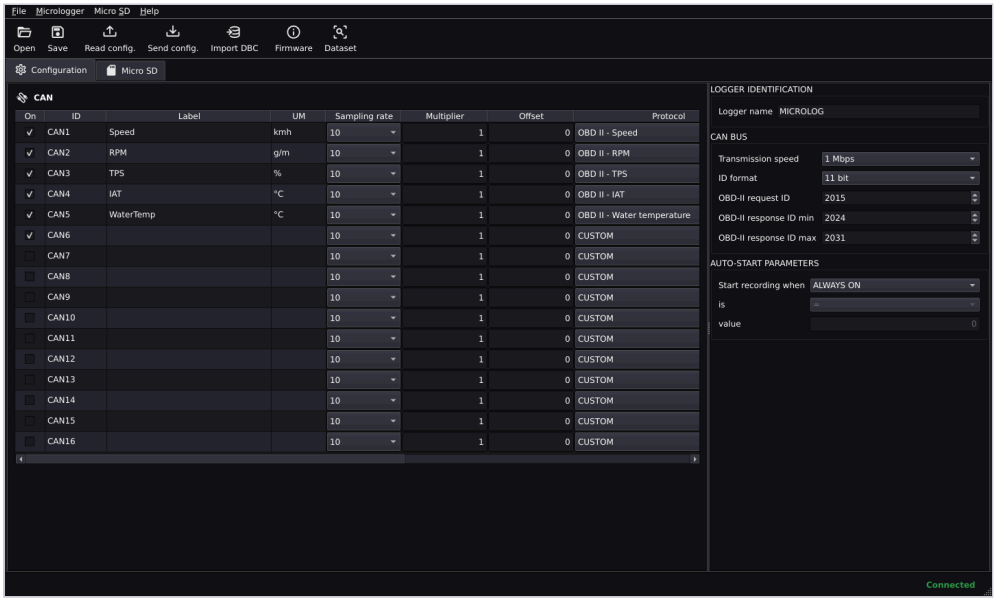


Figure 7 – The Micrologger module: CAN channels, name, CAN bus and auto-start.

5.3 CAN channels

The CAN1-CAN16 table enables each channel to be recorded and sets its label, unit of measurement, frequency, multiplier and offset (value = raw × multiplier + offset).

- **CUSTOM** (Open CAN): CAN ID in decimal, start bit position (0-63), length (1-32 bits), type (signed or unsigned) and format (little or big endian).
- **OBD-II and J1939**: predefined signals (e.g. engine speed, speed, throttle) chosen from a list. The predefined OBD-II IDs suit almost all vehicles.
- **DBC import**: **File → Import DBC…** fills in the channels from the signals of a DBC file.
- **CAN bus**: speed (1 Mbps, 500 kbps, 250 kbps) and ID format (11 or 29 bits).
- **Logger name**: name of the logger, used in the names of the downloaded files.

> **Example: engine speed via OBD-II**
> Enable CAN1, choose the *OBD II - RPM* protocol and a frequency of 10 Hz: MICROLOG requests the engine speed from the ECU ten times per second.

5.4 Auto-start conditions

In **AUTO-START PARAMETERS** you choose *ALWAYS ON* (recording from power-on) or a channel CAN1-CAN16, a condition (=, >, <) and a threshold: the dataset is created when the condition becomes true and is closed at power-off.

> **Example: record only with the engine running**
> Start recording when **CAN1** (engine speed) **> 500**.

5.5 Saving and sending the configuration

1. **File → Save** / **Save as…** saves the configuration on the PC (`.xml` file with the labels alongside in `.gdch`).
2. **Micrologger → Send configuration**: DataStudio assigns an ID to the configuration, so that each dataset gets the right labels; MICROLOG saves it on the microSD and restarts.
3. At each power-on the logger reloads the configuration and creates a new numbered dataset, immediately or when the auto-start condition occurs; the LED changes its blinking when recording is active.

5.6 Datasets and download from the microSD

Micrologger → Datasets in the Micrologger shows the list of recorded datasets via USB. To download them, remove the microSD from the logger and read it with the PC's card reader in the **Micro SD** tab: choose the card (usually the last one in the list), read the datasets and download the one you want, with an automatic name (logger, date, event, session) and optional export to MoTeC or AEM CSV. The `.dat` file is then opened with GUBELLINI DataAnalysis.

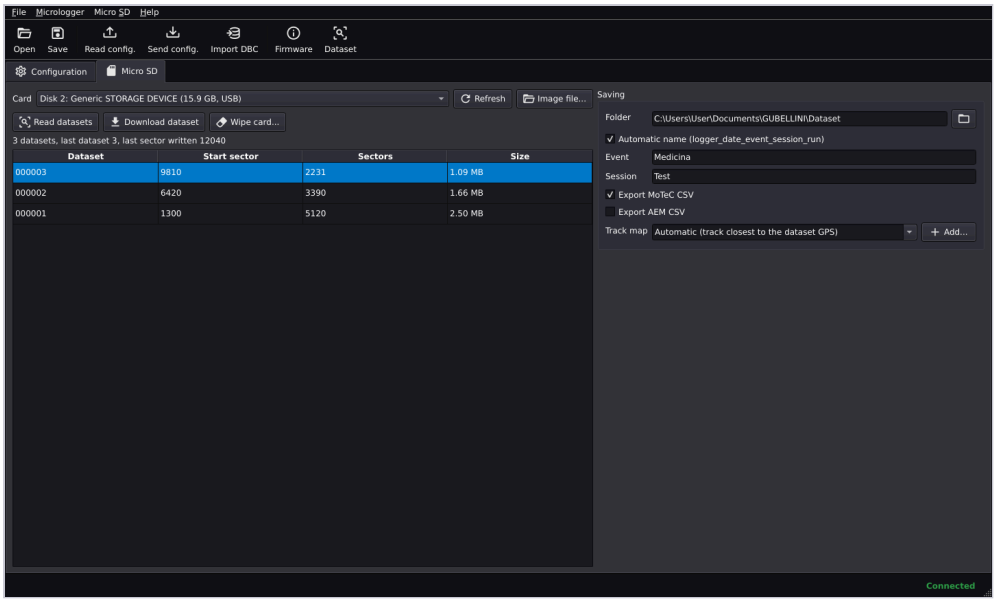


Figure 8 – Micrologger, Micro SD tab.

5.7 AI assistant

The DataStudio **AI assistant** answers questions about MICROLOG connection and configuration, checks the open configuration (for example a disabled auto-start channel, frequencies, CAN parameters) and proposes changes, which the user can apply or reject. It uses a personal API key from an AI provider. To analyse the recorded data, use the GUBELLINI DataAnalysis AI assistant.

6 microSD recording time

The logger records all channels at the frequency of the fastest active channel; each 512-byte sector contains 3 complete samples, regardless of the number of active channels.

Recording frequency	Data per hour	16 GB microSD	32 GB microSD	128 GB microSD
1 Hz	about 0.6 MB	over 25,000 hours	over 50,000 hours	over 200,000 hours
10 Hz	about 6 MB	about 2,600 hours	about 5,200 hours	about 20,800 hours
50 Hz	about 31 MB	about 520 hours	about 1,040 hours	about 4,160 hours
100 Hz	about 61 MB	about 260 hours	about 520 hours	about 2,080 hours
200 Hz	about 123 MB	about 130 hours	about 260 hours	about 1,040 hours

7 Ordering information

7.1 Data loggers

Code	Description
GE-ML-0001	MICROLOG-01 – CAN logger with 16 CAN channels, PA12 + 35% GF housing
On request	MICROLOG-01R – aluminium housing

Package contents: logger, USB-C cable, instruction manual, GUBELLINI DataStudio and GUBELLINI DataAnalysis software.

7.2 CAN modules

Code	Description
GE-GN-0001	GUBENAV – high-dynamics 100 Hz GPS and IMU sensor, on CAN
GE-CB-0001	CAN-BOARD-ACCELEROMETER – 3-axis accelerometer on CAN
GE-CB-0002	CAN-BOARD-GNSS – coordinates, ground speed, date and time on CAN
GE-CB-0003	CAN-BOARD-LOADCELL – load cell on CAN
GE-CB-0004	CAN-BOARD-OPTO – 2 optoisolated digital frequency inputs on CAN
GE-CB-0005	CAN-BOARD-THERMOK – type K thermocouple on CAN

7.3 Related products

Code	Description
GE-GL-0001 / 0003	GUBELOG-01 / GUBELOG-03 – 38-channel data logger with GPS, IMU, analog and digital inputs and CAN

Information and orders
GUBELLINI s.a.s. di Diego Gubellini & C. – Via Euridia Bergianti 10B, 40059 Medicina (BO) Italy
E-mail: info@gubellinielectronics.com · Tel. +39 347 9200208 · WhatsApp +39 380 1978288
www.gubellinielectronics.com

Technical data are subject to change without notice.